APPENDIX I

THE GAMBIA—DEBT SUSTAINABILITY ANALYSIS UNDER THE DEBT SUSTAINABILITY FRAMEWORK FOR LOW-INCOME COUNTRIES—AN UPDATE[26]

1. **This debt sustainability analysis (DSA) updates the last DSA presented to the Fund Board in October 2006.**[27] The DSA is consistent with the HIPC completion point DSA; both DSAs were prepared jointly by staffs of the Fund and the World Bank, and are based on reconciled external debt and debt service data. The full effect of MDRI relief, which was shown as a separate scenario in IMF Country Report No. 06/444, Appendix I is now included in the baseline scenario, as are new borrowing limits under the PRGF-supported program.

2. **The update suggests that The Gambia is at high risk of remaining in debt distress even after receiving HIPC and MDRI relief.** With the full delivery of HIPC and MDRI assistance, the NPV of external debt-to-export ratio declines significantly but remains above the policy-dependent external debt-burden threshold. By contrast, the NPV of debt-to-GDP ratio as well as all the debt service indicators fall below their respective thresholds. Standard stress tests show that The Gambia will remain vulnerable to adverse developments. In particular, the debt stock indicators deteriorate significantly and exceed their respective thresholds when subject to negative shocks. At approximately 30 percent of GDP, the large domestic debt stock further supports the assessment that the risk of debt distress remains high.

I. EVOLUTION OF PUBLIC DEBT SINCE HIPC DECISION POINT

3. **The Gambia reached the Decision Point under the Enhanced HIPC Initiative in December 2000.** Based on external debt data at the end of 1999, the Boards of the International Development Association (IDA) and the IMF approved debt relief worth US$66.6 million in NPV terms.[28] Interim relief in the form of debt service reductions was provided by IDA, the IMF, the African Development Bank (AfDB) and the EU. Reaching the HIPC completion point, originally expected around the end of 2002, was delayed by serious policy slippages which derailed the program supported by the IMF's Poverty Reduction and Growth Facility (PRGF).

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[26] The DSA has been produced jointly by Bank and Fund staffs. The last DSA was presented to the Fund Board in October 2006 (IMF Country Report No. 06/444, Appendix I).

[27] Appendix I in IMF Country Report No. 06/444.

[28] Assuming the participation of all creditors and full delivery of traditional debt relief.

4. **At the end of 2006, the latest date for which comprehensive data are available, the stock of external public debt was US$676.7 million (133.5 percent of GDP)** (Text table 1). Multilateral creditors accounted for 84 percent of this debt, with IDA as the largest creditor (39 percent of total outstanding debt). The nominal debt has grown by US$232 million, or approximately 50 percent (or 30 percentage points of GDP) since end-1999 (point of reference for HIPC decision point in 2000). Most of this increase can be attributed to new disbursements from IDA, the AfDB, other multilateral creditors and non-Paris Club creditors. Since end-2005 (presented in the 2006 DSA), nominal debt has increased by US$49 million, or about 8 percent; most of this increase is the result of new borrowing in 2006, with the remainder due to previously unrecorded debt revealed by the data reconciliation exercise carried out for this DSA.

5. **The stock of gross domestic debt stood at 32.2 percent of GDP at end-2006, up 8 percentage points from its level at decision point.** The increase reflected large extra-budgetary spending in 2001, and the central bank's sale of government debt between 2003 and 2005 to sterilize the impact of rebuilding international reserves.

Text table 1: Evolution of Public External Debt since Decision Point 1/

	1999		2005		2006	
	US$ millions	percent	US$ millions	percent	US$ millions	percent
Total	**444.9**	**100**	**628.2**	**100**	**676.7**	**100**
Multilateral	**355.8**	**80**	**525.2**	**84**	**566.2**	**84**
of which:						
IDA	172.7	39	254.9	41	263.6	39
AfDB	119.3	27	168.4	27	174.5	26
IMF	11.3	3	14.6	2	17.8	3
Other	52.4	12	87.3	14	110.3	16
Bilateral	**89.2**	**20**	**103.1**	**16**	**110.6**	**16**
of which:						
Paris Club	29.8	7	16.0	3	15.6	2
Non-Paris Club	59.3	13	87.1	14	95.0	14
Commercial	**0.0**	**0**	**0.0**	**0**	**0.0**	**0**
Memorandum items:						
GDP (millions US dollars)	**431.9**	...	**461.3**	...	**506.7**	...
Nominal debt-to-GDP	**103.0**	...	**136.2**	...	**133.5**	...

Source: Staff estimates and authorities.

1/ Figures for 1999 are based on reconciled completion point data, 2005 data from IMF Country Report No. 06/444.

II. KEY ASSUMPTIONS

6. **The baseline scenario, consistent with the HIPC completion point DSA, assumes a continuation of recent prudent macroeconomic management, and the country reaching the HIPC-completion point at end-2007.** The key macroeconomic assumptions

are presented in Text table 2. These differ only marginally from those assumptions underlying the last DSA. Notably, real growth has been revised up through 2012, inflation is also projected to be somewhat higher, while export growth over the medium term is more conservative. The revised assumptions reflect:

- Robust output growth over the near term, in line with recent experience, due to on-going large foreign direct investment and donor-sponsored projects. Growth is sustained in the medium term by continued expansion in the tourism and telecommunications sectors, as well as by growth in agriculture.

- Increased exports of goods and services from growth in the tourist sector, as well as groundnuts and other agricultural exports.

- Further growth in imports through 2012, though lower than recent experience. The associated current account deficits due to large externally-financed investment projects decline over the medium term.

- Improved fiscal balance in the medium term, reflecting sustained strong revenue performance, fiscal discipline and lower debt service.

- Increased external donor assistance (both concessional loans and grants) over the period 2007-12, declining as a share of GDP over the medium term.

  - In line with the authorities' PRSP aspirations, external borrowing is expected to average 5.8 percent of GDP between 2007 and 2012 before declining to 2 percent by 2026. New borrowing is expected to be evenly divided between multilateral and bilateral creditors, with an average grant element of 45 percent.

  - External grants increased in 2007 and are expected to average about 4 percent of GDP through 2012 before declining to about 1 percent by 2027.

Text table 2: Key macroeconomic assumptions
(Period averages: in percent, unless otherwise indicated)

	1998-2000	2001-03	2004-06	1998-06	2007-12	2013-27
Real GDP growth	6.1	3.1	6.2	5.2	6.2	5.0
Inflation					4.75	4.0
Growth exports of goods and services	1.1	-0.1	11.6	4.2	5.6	7.7
Non-interest current account (percent of GDP) 1/	-1.5	-2.1	-9.0	-4.2	-10.1	-6.8
Overall fiscal balance (percent of GDP) 1/					-2	-1.7

1/ Including official transfers and grants.

III. External Debt Sustainability

A. Baseline

7. **Under the staff baseline scenario, all but one of the NPV of debt indicators will remain below their corresponding thresholds once completion point is reached and the HIPC/MDRI relief is delivered (Table 1 and Figure 1).**[29] The NPV of debt-to-GDP ratio falls to 26 percent at the beginning of the projection period. New borrowing associated with increased investment pushes the NPV of debt-to-GDP towards the threshold through 2012 before it declines as investment levels off and growth is sustained. The NPV of debt-to-revenue and the debt service ratios fall considerably beneath their respective thresholds. While they too increase through 2012, they remain at comfortable levels throughout the projection period.

Box A1: Policy Dependent Debt-Burden Thresholds for The Gambia

The latest World Bank Country Policy and Institutional Assessment (CPIA) rates The Gambia a "poor performer." The relevant indicative thresholds for countries assessed in this category are 30 percent for the NPV of debt-to-GDP, 100 percent for NPV of debt-to-exports, 200 percent for NPV of debt-to-revenues, 15 percent for debt service–to-exports, and 25 percent for debt service-to-revenues ratio. These ratios are applicable to public and publicly guaranteed external debt.

8. **The NPV of debt-to-exports ratio breaches the debt-burden threshold for a protracted period.** The NPV of debt-to-export ratio is expected to decline significantly to 108 percent in 2007 following full impact of HIPC and MDRI, but this is still above the indicative policy-dependent threshold of 100 percent. Furthermore, this ratio increases through 2012 due to new borrowing, and it stays above the threshold for a protracted period. It gradually falls below the threshold over the medium term, due to sustained growth in receipts from tourism, re-export services, and agricultural exports. While there are country-specific factors to consider when assessing the sustainability of The Gambia's external debt (Box A2), alternative measurements of the ratio continue to indicate a protracted breach of the debt-burden threshold.

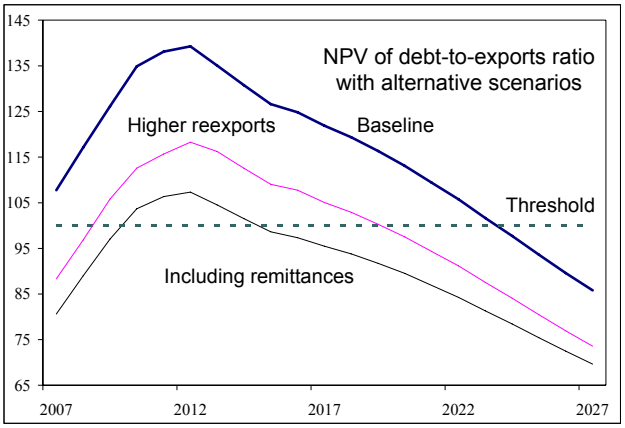
[29] As outlined in the HIPC DSA, outstanding debt at end-2007 before completion point is estimated at NPV US$439 million. Following completion point, HIPC assistance reduces the NPV of existing debt by US$92 million while MDRI results in an additional US$182.1 million reduction.

Box A2: Alternative measures of NPV of debt to exports

The NPV of debt-to-exports threshold is breached over most of the projection period despite HIPC and MDRI relief. The ratio captures the NPV of debt compared with exports of goods and nonfactor services, as an important measure of the country's capacity to sustain external debt levels through its foreign exchange earnings. However, there are country-specific issues to consider when applying this ratio to The Gambia:

- First, official statistics do not capture the full extent of reexport trade and associated service earnings, which are major components of exports. Due to the lack of reliable data, staff estimate re-exports as a fraction of imports, adding a mark-up for earnings from re-exports.
- Second, The Gambia's diaspora consistently remit significant sums (perhaps 10 percent of GDP) which reflect current and prospective foreign exchange earnings (the DSA is based on an empirical analysis that specifically excludes remittances).

The chart below offers a perspective on the possible importance of these factors when assessing the sustainability of The Gambia's external debt. In the chart, the baseline scenario discussed in the text is compared with two alternatives: (i) the case where service exports are revised upward by assuming re-exports of 50 percent of imports rather than 30 percent currently estimated; and, (ii) remittances are added to estimated baseline goods and nonfactor service earnings to reflect projected total foreign exchange earnings in the future. Both effects are seen to drive the NPV of debt to "exports" beneath the threshold in 2007, although this is still breached for an extended period due to new borrowing through 2010 before falling off in the medium term. These alternative measurements substantially improve the outlook for debt sustainability, although not enough to suggest that the country is no longer at a high risk of remaining in debt distress.



B. Alternative Scenarios and Stress Tests

9. **Alternative scenarios reveal that external debt indicators are vulnerable to substantial deterioration in the face of adverse external developments or if new borrowing is on less favorable terms than in the baseline (Table 2, Figure 1).** The scenarios in Figure 1 focus on an "extreme shock" case—reflecting a combined negative

shock to all variables based on historical performance—and if borrowing occurs on less favorable terms.[30] Emphasizing the "less favorable terms" scenario demonstrates the importance of the authorities' commitment to borrowing only on highly concessional terms. With the exception of the debt service indicators, which remain beneath their thresholds, the debt stock indicators all deteriorate significantly under these alternative scenarios. A number of points are worth noting:

- The NPV of debt-to-GDP and the debt-to-exports ratio deteriorate significantly above their respective thresholds (Figure 1 and Table 2). A combination of adverse shocks or lower grant element on debt generates a significant deterioration in debt dynamics over the projection period.

- The debt-to-revenue ratio shows a similar deterioration under the adverse shocks scenario.

- Table 2 (line B6) demonstrates the deterioration in the debt-to-GDP ratio as a result of an exchange rate depreciation. In particular, a one-time 30 percent depreciation results in the debt-to-GDP ratio breaching the 30 percent threshold for much of the projection period. This scenario serves to illustrate the importance of recent exchange rate appreciation on the baseline debt-to-GDP ratio. If this appreciation is not permanent, debt-to-GDP will rise above the indicative threshold in the near term despite HIPC and MDRI relief.

IV. Public Debt Sustainability

A. Baseline

10. **Domestic debt is expected to fall from 29.6 percent of GDP at the end of 2007 to 10.4 percent of GDP in 2012 and to 9.1 percent of GDP in 2027, reflecting sustained good fiscal performance.** Recently implemented reforms in tax administration are expected to help maintain revenues around 21 percent of GDP, while the relatively tight fiscal policy programmed for the medium term helps bring domestic interest rates close to single digits levels. Over the longer term, lower debt service, reflecting the drop in external debt after delivery of debt relief and lower interest rates on domestic debt, provides space to steadily

[30] Figure 1 does not report the standard historical scenario since this shows improved debt dynamics driven by lower historical current account deficits reducing external debt in this scenario. This scenario assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, grace and maturity periods are unchanged, and implies a lower grant element of 25 percent on new borrowing, rather than 45 percent in the baseline.

increase basic primary expenditures[31] and to partially offset the downward trend in externally-financed projects.

11. **The NPV of public debt is projected to decline from about 54.9 percent of GDP in 2007 to 38.9 percent in 2012 and to 30.6 percent in 2027 (Table 3 and Figure 2).** The decline in the near term is driven by a drop in the domestic debt. As a ratio of domestic revenues, the NPV of public debt is projected to fall from about 247 percent in 2007 to 138 percent at the end of the projection period.

B. Alternative Scenarios and Stress Tests

12. **Stress tests indicate that the public debt would not decline significantly under some adverse shocks (Table 4 and Figure 2).** For example, under a permanently lower output growth rate (4.3 percent instead of 5 percent), the NPV of total debt-to-GDP ratio would decline from 54 percent in 2007 to 50 percent in 2027, as opposed to declining to 31 percent under the baseline scenario.

V. Conclusion

13. **It is the staffs' view that The Gambia is at high risk of debt distress based on external debt burden indicators.** The Gambia's debt situation has become more precarious since the last DSA due to the new borrowing. Given continuing risks, development of a comprehensive public debt management strategy should be a priority. High levels of concessionality in new borrowing and careful control over the volume of such borrowing for the foreseeable future are also strongly recommended. Failure to achieve sustained export diversification and growth, meanwhile, would offset any gains in debt management, thus also requiring sustained efforts to promote exports and stimulate growth.

[31] Defined as expenditures excluding interest payments and externally-financed projects.

Table 1. The Gambia: External Debt Sustainability Framework, HIPC and MDRI Scenario, 2007-2027 1/
(In percent of GDP, unless otherwise indicated)

	Actual											Historical Average 6/	Standard Deviation 6/	Projections									
	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) 1/	**...**	**101.8**	**102.5**	**103.0**	**110.6**	**117.3**	**143.0**	**157.3**	**146.5**	**134.7**	**133.6**			**49.9**	**47.5**	**50.7**	**52.7**	**52.6**	**51.7**		**45.6**	**33.9**	
o/w public and publicly guaranteed (PPG)	...	101.8	102.5	103.0	110.6	117.3	143.0	157.3	146.5	134.7	133.6			49.9	47.5	50.7	52.7	52.6	51.7		45.6	33.9	
Change in external debt	...	...	0.7	0.5	7.6	6.7	25.7	14.3	-10.8	-11.8	-1.1			-83.7	-2.4	3.2	2.0	-0.1	-0.9		-1.2	-1.1	
Identified net debt-creating flows	...	...	-4.0	0.1	2.1	0.9	15.7	8.3	-26.8	-17.3	-15.2			-5.7	2.2	1.7	0.8	-0.2	-0.3		0.0	1.9	
Non-interest current account deficit	**...**	**-0.3**	**1.0**	**1.5**	**2.0**	**1.4**	**1.6**	**3.4**	**4.1**	**13.2**	**9.8**	**3.8**	**4.3**	**9.6**	**11.8**	**10.6**	**9.4**	**8.4**	**7.9**	**9.6**	**6.6**	**7.3**	6.8
Deficit in balance of goods and services	...	5.7	7.4	5.8	8.2	5.2	8.4	7.5	21.0	22.7	17.3			19.1	20.3	16.6	15.0	13.0	11.1		9.0	10.2	
Exports	...	27.0	30.1	28.0	26.8	25.8	30.4	31.8	31.7	28.4	30.6			23.9	21.0	20.8	20.3	20.1	19.9		20.7	22.8	
Imports	...	32.7	37.5	33.8	35.0	31.0	38.8	39.3	52.7	51.1	47.9			43.0	41.3	37.3	35.4	33.0	30.9		29.7	33.0	
Net current transfers (negative = inflow)	...	-7.7	-9.3	-7.8	-9.9	-8.3	-11.6	-9.6	-23.6	-17.1	-15.2	-12.0	5.1	-14.7	-13.1	-12.3	-11.8	-11.3	-9.1		-7.9	-5.9	-7.3
o/w official	...	...	...	...	-7.4	-7.5	-10.6	-8.5	-8.5	-5.1	-3.2			-5.2	-5.3	-5.2	-4.9	-4.8	-2.8		-2.5	-1.9	
Other current account flows (negative = net inflow)	...	1.6	2.9	3.5	3.7	4.6	4.7	5.4	6.7	7.5	7.8			5.2	4.7	6.3	6.1	6.7	5.9		5.5	3.0	
Net FDI (negative = inflow)	**...**	**-3.8**	**-3.7**	**-0.1**	**-3.8**	**-2.4**	**-2.5**	**-3.6**	**-14.1**	**-13.3**	**-14.6**	**-7.8**	**5.5**	**-9.2**	**-7.4**	**-6.4**	**-6.2**	**-6.0**	**-5.9**		**-4.8**	**-4.0**	-4.6
Endogenous debt dynamics 2/	**...**	**...**	**-1.3**	**-1.3**	**3.9**	**2.0**	**16.5**	**8.5**	**-16.8**	**-17.2**	**-10.4**			**-6.1**	**-2.3**	**-2.5**	**-2.4**	**-2.5**	**-2.3**		**-1.8**	**-1.4**	
Contribution from nominal interest rate	...	...	1.4	1.4	1.2	1.2	1.2	1.7	2.0	1.9	1.7			1.2	0.4	0.4	0.4	0.4	0.4		0.4	0.3	
Contribution from real GDP growth	...	...	-6.4	-6.4	-5.8	-6.4	4.3	-10.3	-9.7	-6.4	-8.0			-7.3	-2.6	-2.8	-2.8	-2.9	-2.7		-2.2	-1.6	
Contribution from price and exchange rate changes	...	...	3.8	3.8	8.5	7.2	11.0	17.1	-9.1	-12.7	-4.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**0.0**	**...**	**4.7**	**0.4**	**5.5**	**5.8**	**10.0**	**6.0**	**16.0**	**5.5**	**14.0**			**-78.0**	**-4.6**	**1.5**	**1.2**	**0.1**	**-0.6**		**-1.3**	**-3.0**	
o/w exceptional financing	...	...	0.0	0.0	0.0	0.0	-0.1	0.0	-0.4	-0.3	-0.3			-58.4	-0.9	-0.7	-0.6	-0.4	-0.4		-0.3	-0.1	
NPV of external debt 4/	...	...	...	...	...	...	...	...	...	...	31.1			25.5	24.8	27.1	28.7	29.2	29.2		26.9	21.6	
In percent of exports	...	...	...	...	...	...	...	...	...	...	101.7			106.8	118.5	130.6	141.5	145.7	146.9		130.2	94.7	
NPV of PPG external debt	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**31.1**			**25.5**	**24.8**	**27.1**	**28.7**	**29.2**	**29.2**		**26.9**	**21.6**	
In percent of exports	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**101.7**			**106.8**	**118.5**	**130.6**	**141.5**	**145.7**	**146.9**		**130.2**	**94.7**	
In percent of government revenues	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**146.8**			**120.9**	**119.6**	**133.5**	**142.8**	**147.4**	**147.6**		**135.0**	**106.3**	
Debt service-to-exports ratio (in percent)	**...**	**18.0**	**16.0**	**...**	**19.6**	**16.8**	**23.9**	**11.1**	**17.7**	**17.7**	**16.5**			**18.5**	**7.0**	**7.0**	**6.8**	**6.9**	**7.6**		**8.3**	**6.4**	
PPG debt service-to-exports ratio (in percent)	**...**	**-14.3**	**-7.7**	**...**	**19.6**	**16.8**	**23.9**	**11.1**	**17.7**	**17.7**	**16.5**			**18.5**	**7.0**	**7.0**	**6.8**	**6.9**	**7.6**		**8.3**	**6.4**	
PPG debt service-to-revenue ratio (in percent)	**...**	**-21.0**	**-13.3**	**...**	**28.4**	**28.8**	**44.5**	**22.4**	**26.9**	**25.5**	**23.8**			**20.9**	**7.1**	**7.1**	**6.8**	**6.9**	**7.6**		**8.6**	**7.1**	
Total gross financing need (millions of U.S. dollars)	...	...	...	...	14.5	13.9	23.6	11.7	-17.6	22.8	1.2			27.3	47.2	48.1	42.5	37.1	37.4		53.2	136.8	
Non-interest current account deficit that stabilizes debt ratio	...	...	0.3	0.9	-5.7	-5.3	-24.1	-10.9	14.9	25.0	11.0			93.2	14.2	7.4	7.4	8.4	8.8		7.9	8.4	
Key macroeconomic assumptions																							
Real GDP growth (in percent)	...	...	6.5	6.4	5.5	5.8	-3.2	6.9	7.0	5.1	6.5	5.2	3.2	7.0	6.5	6.5	6.0	6.0	5.5	6.2	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	...	...	-3.6	-3.5	-7.7	-6.1	-8.6	-10.7	6.1	9.5	3.1	-2.4	7.0	19.6	15.0	1.9	1.8	2.0	1.8	7.0	1.8	1.8	1.7
Effective interest rate (percent) 5/	...	...	1.4	1.4	1.1	1.1	0.9	1.1	1.5	1.5	1.4	1.3	0.2	1.2	0.9	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.8
Growth of exports of G&S (US dollar terms, in percent)	...	...	14.5	-4.2	-7.0	-4.3	4.3	-0.3	13.3	2.9	18.4	4.2	9.2	-0.1	7.5	7.4	5.6	6.8	6.5	5.6	8.1	8.1	7.8
Growth of imports of G&S (US dollar terms, in percent)	...	...	17.6	-7.4	0.9	-12.0	10.8	-3.3	52.3	11.6	2.9	8.1	19.1	14.9	17.5	-1.9	2.3	1.0	0.7	5.7	7.3	8.1	7.3
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	...	...	...	...	...	...	...	42.6	43.2	43.3	44.1	45.0	45.0	43.9	45.0	45.0	45.0
Aid flows (in millions of US dollars) 7/	0.0	19.7	22.4	18.5	25.5	33.3	51.4	35.1	50.6	41.5	44.2			60.8	88.0	98.9	94.8	84.2	75.0		76.0	87.0	
o/w Grants	5.5	8.2	4.8	...	10.5	16.5	12.4	...	18.2	7.7	0.3			24.2	31.6	36.2	36.1	35.4	33.8		29.8	20.5	
o/w Concessional loans	0.0	14.2	...	...	...	...	...	...	32.3	33.8	43.9			36.5	56.5	62.8	58.7	48.8	41.2		46.2	66.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...	...	...	...	...	...	...	...	...			6.9	7.4	7.7	6.8	5.7	4.8		3.4	1.7	2.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...	...	...	...	...	...	...	...	...			61.8	62.2	62.9	64.7	68.1	69.8		66.6	58.0	64.3
Memorandum items:																							
Nominal GDP (millions of US dollars)	...	409.8	420.8	431.9	420.9	417.9	369.7	353.0	401.0	461.3	506.7			648.4	793.8	861.4	929.4	1005.1	1079.8		1492.3	2898.2	
(NPVt-NPVt-1)/GDPt-1 (in percent)														1.5	4.9	4.6	3.9	2.9	2.1	3.3	1.4	0.9	1.2

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Table 2. The Gambia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
including HIPC and MDRI (In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of debt-to-GDP ratio								
Baseline	25	25	27	29	29	29	**27**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	25	25	26	26	25	24	**16**	6
A2. New public sector loans on less favorable terms in 2008-27 2/	25	26	30	33	35	35	**36**	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	25	26	30	31	32	32	**29**	24
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	25	26	31	32	33	33	**29**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	25	32	39	41	42	42	**38**	31
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	25	32	40	41	41	41	**35**	25
B5. Combination of B1-B4 using one-half standard deviation shocks	25	36	49	51	51	51	**45**	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	25	34	37	39	40	39	**36**	29
NPV of debt-to-exports ratio								
Baseline	107	118	131	142	146	147	**130**	95
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	107	121	123	126	124	120	**80**	25
A2. New public sector loans on less favorable terms in 2007-26 2/	107	126	146	164	173	178	**173**	146
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	107	118	131	142	146	147	**130**	95
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	107	141	191	204	209	210	**182**	126
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	107	118	131	142	146	147	**130**	95
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	107	151	191	201	204	204	**170**	108
B5. Combination of B1-B4 using one-half standard deviation shocks	107	147	198	210	214	214	**182**	120
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	107	118	131	142	146	147	**130**	95
NPV of debt-to-revenue ratio								
Baseline	121	120	133	143	147	148	**135**	106
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	121	122	126	127	126	120	**83**	28
A2. New public sector loans on less favorable terms in 2007-26 2/	121	128	150	166	175	179	**180**	163
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	121	125	146	156	161	161	**147**	116
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	121	126	152	161	165	165	**148**	111
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	121	152	190	204	210	211	**193**	152
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	121	153	196	203	207	205	**176**	121
B5. Combination of B1-B4 using one-half standard deviation shocks	121	173	243	254	260	258	**226**	162
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	121	162	181	193	200	200	**183**	144

Table 2. The Gambia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27 (continued)
with HIPC and MDRI (In percent)

	Projections							
	2007	2008	2009	2010	2011	2012	**2017**	2027
Debt service-to-exports ratio								
Baseline	18	7	7	7	7	8	**8**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	19	9	9	8	8	10	**8**	4
A2. New public sector loans on less favorable terms in 2008-27 2/	19	7	8	8	9	9	**11**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	19	7	7	7	7	8	**8**	7
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	19	8	9	9	9	10	**12**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	19	7	7	7	7	8	**8**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	19	7	8	8	8	8	**12**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	19	8	9	9	9	9	**12**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	19	7	7	7	7	8	**8**	7
Debt service-to-revenue ratio								
Baseline	21	7	7	7	7	8	**9**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	21	9	9	8	9	10	**8**	4
A2. New public sector loans on less favorable terms in 2008-27 2/	21	7	8	8	9	9	**11**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	21	8	8	7	8	8	**9**	8
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	21	7	7	7	7	8	**10**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	21	9	10	10	10	11	**12**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	21	7	8	8	8	8	**12**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	21	9	11	10	11	11	**15**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	21	10	10	9	9	10	**12**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	45	45	45	45	45	45	**45**	45

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.The Gambia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-2027
(In percent of GDP, unless otherwise indicated)

	Actual								Estimate	Projections								
	2001	2002	2003	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt 1/	155.4	179.6	185.2	179.4	170.2	165.5			79.1	71.3	70.8	68.3	64.8	60.9		53.9	42.9	
o/w foreign-currency denominated	117.3	143.0	157.3	146.5	134.7	133.3			49.5	47.0	50.0	51.7	51.5	50.5		44.0	33.8	
Change in public sector debt	13.3	24.2	5.6	-5.8	-9.2	-4.7			-86.4	-7.8	-0.5	-2.5	-3.5	-3.9		-1.2	0.2	
Identified debt-creating flows	-14.0	27.4	-8.5	-30.8	-14.3	-5.3			-44.5	-6.5	-4.3	-5.4	-5.1	-5.3		-2.3	-1.1	
Primary deficit	9.4	1.5	0.0	-0.1	0.4	-0.6	0.7	3.4	-5.5	0.2	0.7	-0.4	-0.3	-0.7	-1.0	1.2	-1.1	0.3
Revenue and grants	16.1	19.1	17.0	24.2	21.3	22.9			25.5	24.8	25.2	24.8	24.5	24.0		22.9	22.7	
of which : grants	1.0	2.8	1.3	3.3	1.5	1.7			3.5	3.7	4.0	3.7	3.4	3.0		1.9	0.7	
Primary (noninterest) expenditure	25.6	20.6	17.0	24.1	21.7	22.3			20.1	25.0	25.9	24.4	24.1	23.4		24.1	21.6	
Automatic debt dynamics	-23.4	28.3	-7.1	-29.3	-14.0	-6.7			-39.2	-3.6	-3.4	-3.4	-3.5	-3.3		-2.5	0.5	
Contribution from interest rate/growth differential	-24.1	3.5	-14.3	-10.6	-4.6	-5.9			-9.1	-3.2	-3.1	-3.2	-3.4	-3.1		-2.5	-1.1	
of which : contribution from average real interest rate	-1.3	-1.7	-2.8	1.5	4.2	4.6			1.7	1.6	1.2	0.8	0.5	0.3		0.1	0.9	
of which : contribution from real GDP growth	-7.7	5.2	-11.5	-12.2	-8.7	-10.5			-10.8	-4.8	-4.3	-4.0	-3.9	-3.4		-2.6	-2.0	
Contribution from real exchange rate depreciation	0.7	24.9	7.2	-18.7	-9.4	-0.8			-30.0	-0.4	-0.3	-0.2	-0.1	-0.2		...	...	
Other identified debt-creating flows	0.0	-2.5	-1.4	-1.5	-0.7	2.0			0.2	-3.1	-1.7	-1.5	-1.3	-1.3		-1.0	-0.4	
Privatization receipts (negative)	0.0	0.0	0.0	0.0	-0.4	0.0			-2.3	-2.2	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	1.6			2.7	1.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-1.0	-2.0	-1.1	-1.3	-0.1	0.0			-0.6	-2.6	-2.4	-2.2	-1.7	-1.6		-1.0	-0.4	
Other (specify, e.g. bank recapitalization)	1.0	-0.4	-0.2	-0.2	-0.2	0.4			0.4	0.6	0.7	0.6	0.4	0.3		0.0	0.0	
Residual, including asset changes	27.3	-3.2	14.1	25.0	5.1	0.6			-41.9	-1.3	3.9	2.9	1.7	1.4		1.1	1.3	
NPV of public sector debt	38.1	36.6	27.9	32.9	35.5	63.3			54.9	48.9	47.5	44.8	41.9	38.9		35.9	30.6	
o/w foreign-currency denominated	0.0	0.0	0.0	0.0	0.0	31.2			25.3	24.6	26.7	28.2	28.6	28.5		26.0	21.5	
o/w external	...	...	...	...	...	31.2			25.3	24.6	26.7	28.2	28.6	28.5		26.0	21.5	
NPV of contingent liabilities (not included in public sector debt)	...	...	...	...	...	...			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 2/	17.1	10.4	6.5	9.3	12.3	10.1			0.6	5.2	5.3	3.7	3.0	2.3		3.8	1.5	
NPV of public sector debt-to-revenue and grants ratio (in percent)	236.0	191.5	164.0	135.8	166.6	276.8			215.0	196.9	188.9	180.3	171.2	161.8		156.7	134.3	
NPV of public sector debt-to-revenue ratio (in percent)	252.2	224.2	177.8	157.2	179.6	298.8			248.6	231.9	224.8	211.8	198.9	184.9		170.7	138.3	
o/w external 3/						147.1			114.6	116.7	126.5	133.5	135.9	135.3		123.7	97.2	
Debt service-to-revenue and grants ratio (in percent) 4/	47.5	57.9	46.4	44.6	54.9	42.9			31.5	18.5	16.2	14.4	12.5	11.8		11.5	11.6	
Debt service-to-revenue ratio (in percent) 4/	50.7	67.8	50.2	51.6	59.2	46.3			36.4	21.8	19.2	17.0	14.6	13.4		12.6	11.9	
Primary deficit that stabilizes the debt-to-GDP ratio				5.8	9.6	4.1			80.9	8.0	1.2	2.1	3.1	3.2		2.4	-1.4	
Key macroeconomic and fiscal assumptions																		
Real GDP growth (in percent)	5.8	-3.2	6.9	7.0	5.1	6.5	5.0	3.1	7.0	6.5	6.5	6.0	6.0	5.5	6.3	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	1.3	1.2	1.6	1.6	1.4	1.4	1.4	0.1	1.0	0.9	1.0	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Average real interest rate on domestic currency debt (in percent)	-2.1	-4.2	-8.7	8.9	17.3	12.9	7.4	9.2	7.4	6.3	6.7	6.9	6.7	6.8	6.8	6.5	13.9	7.7
Real exchange rate depreciation (in percent, + indicates depreciation)	0.7	20.5	5.4	-12.8	-6.8	-0.6	4.9	11.7	-24.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	15.2	16.1	27.4	12.2	4.1	1.4	8.9	8.6	5.0	5.0	5.0	4.8	4.5	4.3	4.8	4.0	-1.0	3.7
Growth of real primary spending (deflated by GDP deflator, in percent)	47.8	-22.1	-11.6	51.6	-5.5	9.3	5.3	25.5	-3.6	32.7	10.3	-0.1	4.8	2.1	7.7	3.0	6.5	4.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	...			42.6	43.2	43.3	44.1	45.0	45.0	43.9	45.0	45.0	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

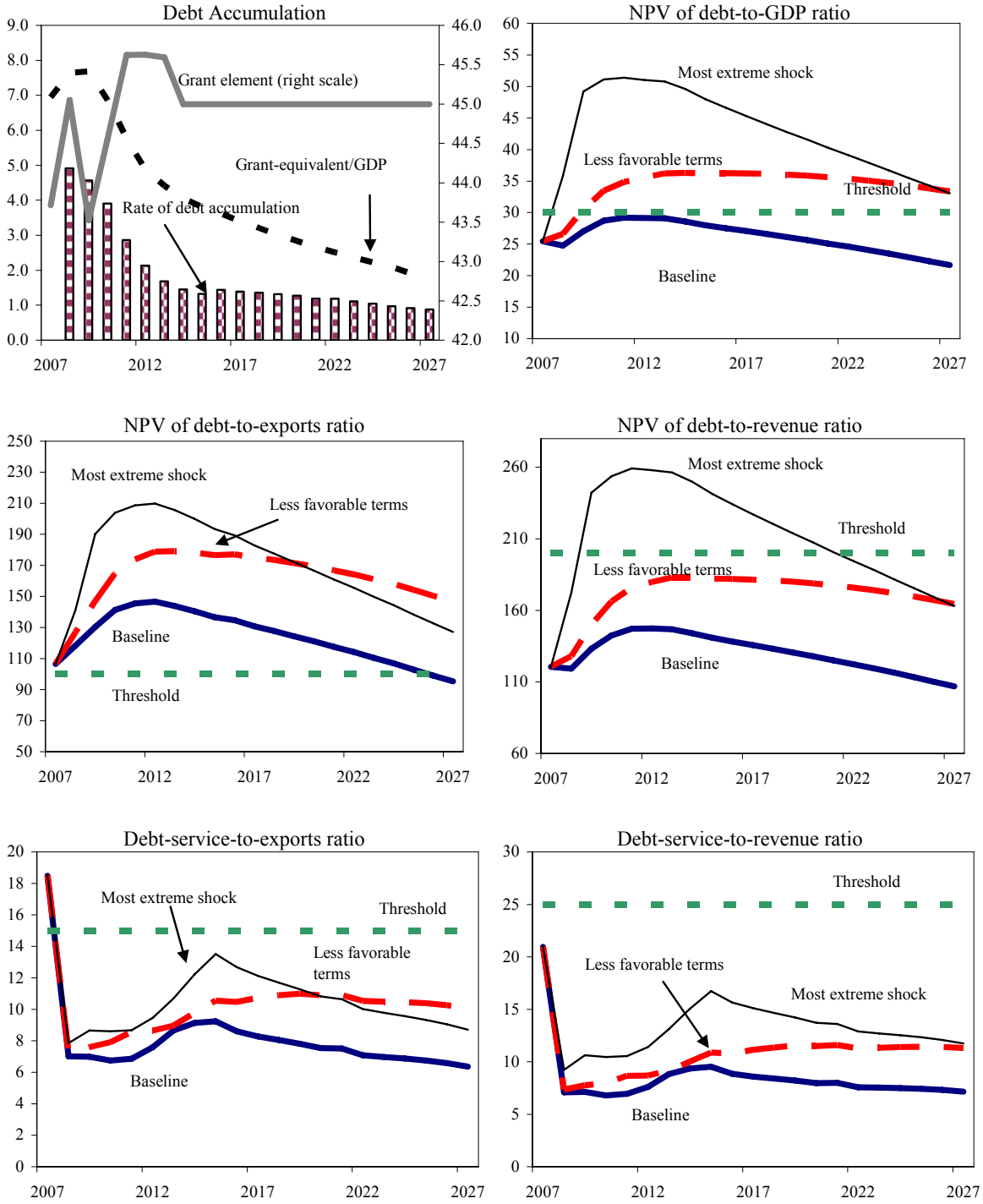
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.The Gambia: Sensitivity Analysis for Key Indicators of Public Debt 2007-2027

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	55	49	48	45	42	39	36	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	54	49	49	48	46	44	39	37
A2. Primary balance is unchanged from 2007	54	45	40	35	30	25	8	-16
A3. Permanently lower GDP growth 1/	54	49	48	46	44	42	43	50
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	54	51	54	52	50	48	49	50
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	54	51	52	49	46	42	39	32
B3. Combination of B1-B2 using one half standard deviation shocks	54	51	52	49	46	43	39	33
B4. One-time 30 percent real depreciation in 2008	54	59	56	52	49	45	41	34
B5. 10 percent of GDP increase in other debt-creating flows in 2008	54	54	52	49	46	43	40	33
NPV of Debt-to-Revenue Ratio 2/								
Baseline	215	197	189	180	171	162	157	134
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	213	199	194	190	186	180	170	163
A2. Primary balance is unchanged from 2007	213	180	160	142	124	105	33	-71
A3. Permanently lower GDP growth 1/	213	197	191	186	179	173	186	219
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	213	205	210	206	201	196	211	217
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	213	205	205	195	186	177	171	142
B3. Combination of B1-B2 using one half standard deviation shocks	213	206	206	197	187	178	170	143
B4. One-time 30 percent real depreciation in 2008	213	239	224	211	199	188	177	148
B5. 10 percent of GDP increase in other debt-creating flows in 2008	213	219	207	198	189	180	174	144
Debt Service-to-Revenue Ratio 2/								
Baseline	31	18	16	14	13	12	12	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	31	21	17	14	14	14	11	15
A2. Primary balance is unchanged from 2007	31	21	9	4	2	1	-4	-9
A3. Permanently lower GDP growth 1/	31	21	17	15	13	12	15	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	31	22	19	18	16	16	17	20
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	31	21	21	20	14	12	12	12
B3. Combination of B1-B2 using one half standard deviation shocks	31	21	20	18	13	12	12	12
B4. One-time 30 percent real depreciation in 2008	31	21	17	15	13	12	13	13
B5. 10 percent of GDP increase in other debt-creating flows in 2008	31	21	29	18	13	12	12	13

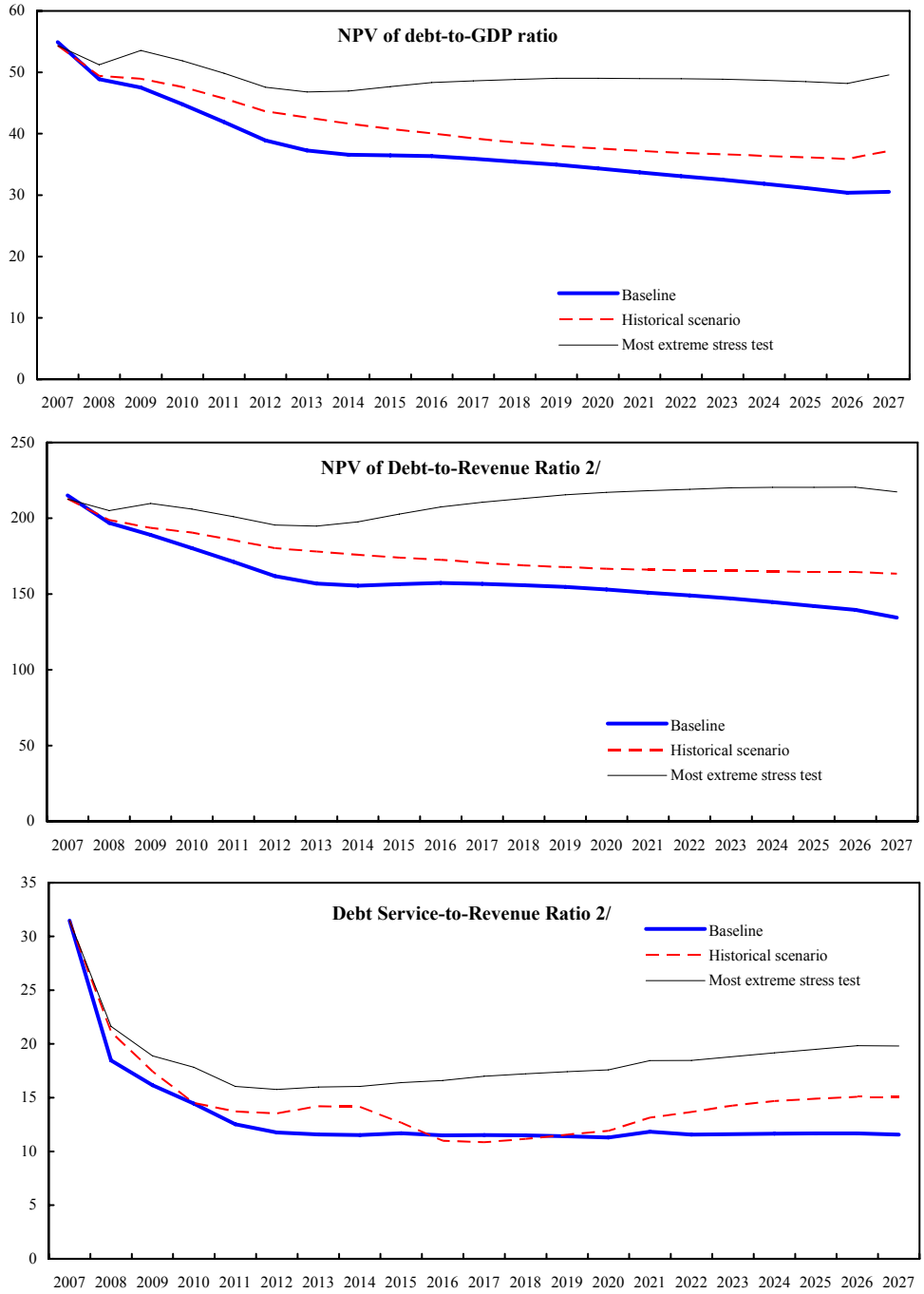
Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure 1. The Gambia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2007-2027



Source: Staff projections and simulations.

Figure 2.The Gambia: Indicators of Public Debt Under Alternative Scenarios, 2007-2027 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017.
2/ Revenue including grants.